Shrinkage priors, admissibility, and weighted Riemannian geometry: a dictionary with status tags

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Two literatures touch without citing each other: Stein-type shrinkage and admissibility (Stein, Brown, Komaki) and the potential theory of weighted Riemannian manifolds (Grigor'yan; Perelman's weighted scalar curvature). This note records the contact points, each with a status tag. (i) Komaki's $O(N^{-2})$ risk reduction of a prior $e^{-\Phi}\pi_J$ over the Jeffreys prior equals $(R^{\mathrm{m}}_{\Phi}-R)/(2N^2)$, the Φ-dependent part of Perelman's weighted scalar curvature; the R term has no statistical counterpart. (ii) Brown's 1971 admissibility theory is the potential theory of the weighted Euclidean space $(\mathbb{R}^m, \delta, f^*\,\mathrm{d}x)$: his Theorem 5.1.1 reads "admissible if and only if parabolic", and his 1971 sphere-average test is the sphere-area parabolicity test that the geometric literature attributes to Lyons–Sullivan (1984) and Grigor'yan (1985). (iii) Two elementary lemmas locate the cases that Brown's radial tests leave open: a bounded spherical A_2 ratio makes his two tests exhaustive, and for product weights the doubling and Poincaré hypotheses of heat-kernel theory force that bound, so a volume-growth criterion decides nothing new there. An explicit undecided example is settled (inadmissible) by product heat kernels, an instance of Johnstone–Lalley (1984). No dynamical and no gravitational claim is made.

1 Scope and status

This is a research note of record, written to close a question rather than to open a field. It replaces an LLM-drafted blueprint that read the same material as "emergent spacetime"; twenty-two machine checks found six of that blueprint's displayed equations wrong, and the present text keeps only what survived against the primary sources. Every claim carries one of four tags. [THEOREM]: stated and proved in the cited source, which we read. [IDENTITY]: an elementary identity, re-derived symbolically in the check scripts. [NEGATIVE]: a conjecture of the blueprint refuted from a source. [OBSERVATION]: sourced on both sides, novelty not established. For novelty the only permitted wording is "not found in the literature we searched"; a citation-index sweep has not been run.

Sources read in full or in the cited parts: Brown [1], Komaki [2, 3], Perelman [4] (§1), Grigor'yan [5] (§§3, 5, 7, 11), Johnstone [6], George–Liang–Xu [7], Maruyama–Strawderman [8], Takanashi–McAlinn [9]. Works marked "not accessed" in the bibliography are quoted as recalled; nothing below depends on them except the heat-kernel remark of §5, which is explicitly conditional, and the Poincaré column of Table 2, which uses Muckenhoupt's weighted Hardy inequality.

2 Setting

Model side. A regular model $\{p(x \mid \theta) : \theta \in \Theta \subset \mathbb{R}^d\}$ carries the Fisher metric $g_{ij} = E[\partial_i \ell\, \partial_j \ell]$, the Jeffreys prior $\pi_J \propto |g|^{1/2}$ and the Laplace–Beltrami operator Δ_g [10]. A smooth prior is written $f = e^{-\Phi}\pi_J$, and $u = \sqrt{f/\pi_J} = e^{-\Phi/2}$. The pair (g, Φ) defines the smooth metric measure space $(M, g, e^{-\Phi}\,\mathrm{d}V)$ with weighted Laplacian $\Delta_\Phi = \Delta_g - \nabla\Phi\cdot\nabla$, self-adjoint on $L^2(e^{-\Phi}\,\mathrm{d}V)$, Bakry–Émery tensor $\mathrm{Ric} + \nabla\nabla\Phi$ [11], and Perelman's weighted scalar curvature [4]

$$R^{\mathrm{m}}_{\Phi} = R + 2\Delta_g\Phi - |\nabla\Phi|^2_g, \tag{1}$$

whose integral is his functional $\mathcal{F}(g,\Phi) = \int R^{\mathrm{m}}_{\Phi}\, e^{-\Phi}\,\mathrm{d}V = \int (R + |\nabla\Phi|^2) e^{-\Phi}\,\mathrm{d}V$. A weighted manifold is *parabolic* if its weighted Brownian

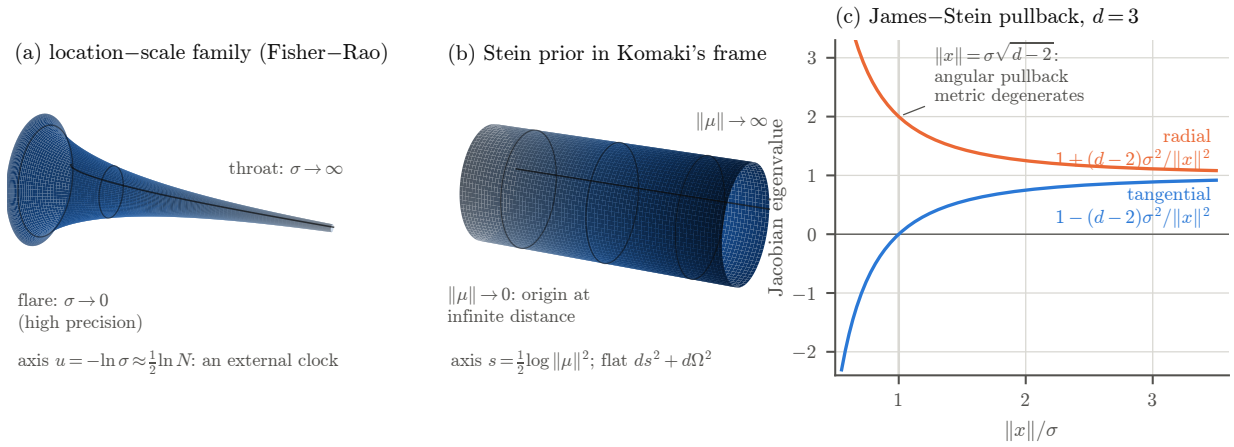


Figure 1: Three of the four geometries of §6. (a) The Fisher–Rao metric of a location–scale family, hyperbolic, drawn as the pseudosphere (curvature -1 after rescaling): throat $\sigma \to \infty$, flare $\sigma \to 0$, axial coordinate $-\ln\sigma \approx \frac{1}{2}\ln N$, an external clock; the meridian is a geodesic. (b) Komaki's conformal frame of the Stein prior: the flat cylinder $S^{d-1} \times \mathbb{R}$ (sphere drawn as a circle) with the origin pushed to infinite distance; constant cross-section, no horizon. (c) Jacobian eigenvalues of the James–Stein map for $d = 3$: the tangential eigenvalue vanishes at $\|x\| = \sigma\sqrt{d-2}$, where the estimator's pullback metric degenerates in the angular directions. Shading encodes the axial coordinate.

motion is recurrent, equivalently if it carries no Green function and no nonconstant positive superharmonic function [5, §5]. $\mathrm{Risk}_N(f)$ is the Kullback–Leibler risk of the Bayesian predictive density built from N observations under the prior f.

Brown's side. $X \sim N_m(\theta, I)$, F a σ-finite prior on $\mathbb{R}^m$, $f^* = p * F$ its marginal (here p is the normal density), and $\delta_F = x + \nabla \log f^* = x + \gamma_F$ the generalized Bayes estimator; δ_F has bounded risk when γ_F is bounded [1, §3]. We write $W_F = (\mathbb{R}^m, \delta_{ij}, f^*\,\mathrm{d}x)$ for the weighted Euclidean space. Brown's diffusion $\{Z_t\}$, with generator $\Delta + \nabla \log f^* \cdot \nabla$ up to a constant factor, is the weighted Brownian motion of W_F. For $r \geq 1$ let $\bar{f}(r)$ and $A(r)$ be the means of f^* and of $1/f^*$ over the sphere $\{|x| = r\}$, and $Q(r) = \bar{f}(r)A(r) \geq 1$ the *spherical A_2 ratio*.

3 Shrinkage priors and the weighted scalar curvature

Proposition 1 ([IDENTITY] on [THEOREM] inputs)**.** *For a regular model and $f = e^{-\Phi}\pi_J$, the risk reduction $\varrho_N(f) = \mathrm{Risk}_N(\pi_J) - \mathrm{Risk}_N(f)$ satisfies*

$$\begin{aligned}\varrho_N(f) &= \frac{1}{2N^2}\left(2\Delta_g\Phi - |\nabla\Phi|_g^2\right) + o(N^{-2}) \\ &= \frac{R^{\mathrm{m}}_\Phi - R}{2N^2} + o(N^{-2}) \\ &= -\frac{2}{N^2}\,\frac{\Delta_g u}{u} + o(N^{-2}). \qquad (2)\end{aligned}$$

The prior f asymptotically dominates π_J if and only if u is a nonconstant positive superharmonic function on (M, g), that is, if and only if $R^{\mathrm{m}}_\Phi \geq R$ with strict inequality somewhere.

Proof. The first equality is Komaki's expansion [2, eqs. (3)–(5)]: in the difference of two priors the skewness terms and the prior-independent terms cancel. The second is the definition (1) with $f \equiv \Phi$ [4, Remark 1.3]; the third is the identity $2\Delta\Phi - |\nabla\Phi|^2 = -4\Delta u/u$ for $u = e^{-\Phi/2}$. The domination statement is Komaki's Theorem 2. (The blueprint had the sign of the first line reversed; checks C8–C10.) □

Proposition 2 ([NEGATIVE])**.** *The prior-independent $O(N^{-2})$ term of the Jeffreys predictive risk is not proportional to the scalar curvature R of (M, g), and $\mathcal{F}$ is not the integrated predictive risk. Whenever the boundary term of one*

integration by parts vanishes, up to $o(N^{-2})$,

$$\int \varrho_N(f)\, e^{-\Phi}\,\mathrm{d}V = \frac{1}{2N^2}\int |\nabla\Phi|_g^2\, e^{-\Phi}\,\mathrm{d}V = \frac{1}{2N^2}\Big[\mathcal{F}(g,\Phi) - \int R\, e^{-\Phi}\,\mathrm{d}V\Big]. \quad (3)$$

Proof. Komaki [3, Theorem A1, eq. (29)] writes the full $O(N^{-2})$ risk. Its prior-independent part is a fixed combination of the exponential and mixture embedding curvatures, the skewness norm $T_{ijk}T^{ijk}$, a fourth-moment contraction and one mixed moment. For $d = 1$ these are generically nonzero (Efron's statistical curvature of a curved exponential family [12]), while every intrinsic curvature of a one-dimensional Riemannian manifold vanishes. Equation (3) follows from (2) and $\int \Delta\Phi\, e^{-\Phi}\,\mathrm{d}V = \int |\nabla\Phi|^2 e^{-\Phi}\,\mathrm{d}V$. □

So only the gradient part of $\mathcal{F}$ is statistical, and the operator that governs shrinkage domination is $-4\Delta_g$ acting on u: Perelman's operator $-4\Delta + R$ with the potential removed. His eigenvalue $\lambda(g)$ has no reading here.

Remark 3 ([THEOREM], Komaki). Komaki's Theorems 1–2 [2] say that a prior asymptotically dominating π_J exists if and only if (M, g) carries a Green function, i.e. is non-parabolic; his Theorem 3 obtains this from negative sectional curvature for $d \geq 3$ via Aomoto [13]. For the flat normal location model the Green function exists iff $d \geq 3$ and the Green prior is Stein's harmonic prior $\|\mu\|^{-(d-2)}$ (Komaki's Example 1). Curvature enters statistics here, through transience, and not through R in the risk.

Proposition 4 ([THEOREM] + [IDENTITY])**.** *For* $d \geq 3$ *and a vague prior* $h = e^{-\Phi}\pi_J$*, Komaki [2, §3, Theorems 4–6] decides domination over* h *by superharmonicity on the conformal metric* $\tilde{g} = (h/\pi_J)^{2/(d-2)} g = e^{-2\Phi/(d-2)} g$*. In that frame, up to boundary terms,*

$$\mathcal{F}(g,\Phi) = \int \Big[\tilde{R} - \tfrac{1}{d-2}\,|\tilde{\nabla}\Phi|_{\tilde{g}}^2\Big]\,\mathrm{d}\tilde{V}, \quad (4)$$

(check C2; the blueprint's coefficient $(d-1)/(d-2)$ *was wrong). For the Stein prior the frame is the flat cylinder* $S^{d-1} \times \mathbb{R}$ *under* $\mu \mapsto (\mu/\|\mu\|, \tfrac{1}{2}\log\|\mu\|^2)$*, with the origin at infinite distance (Komaki's Example 1; Fig. 1b).*

In words: changing the vague prior is a conformal change of the Fisher metric. The "dilaton" form of (4) is a name for a static Riemannian functional and carries no further content.

4 Brown (1971) as potential theory on W_F

Theorem 5 (Brown [1], restated; [THEOREM] on both sides)**.**

(a) (Thm. 4.3.1) $\{Z_t\}$ is transient iff the infimum of $\int |\nabla j|^2 f^*\,\mathrm{d}x$ over compactly supported j with $j = 1$ on $|x| \leq 1$ is positive, i.e. iff $\mathrm{cap}_{W_F}(B_1) > 0$: non-parabolicity of W_F [5, Thm. 5.1(6)].

(b) (Thm. 5.1.1) If γ_F is bounded, δ_F is admissible iff W_F is parabolic; transience implies inadmissibility with no condition.

(c) (Thm. 6.3.1) For spherically symmetric F, δ_F is inadmissible iff $\int_1^\infty \mathrm{d}r/(r^{m-1} f^*(r)) < \infty$: the model-manifold test $\int^\infty \mathrm{d}r/S(r)$ [5, Prop. 3.1] for the weighted sphere area $S(r) = \omega_{m-1} r^{m-1} f^*(r)$ (check C23).

(d) (Thm. 6.4.3) For any F, $\int_1^\infty \mathrm{d}r/(r^{m-1}\bar{f}(r)) = \infty$ implies that W_F is parabolic, hence δ_F admissible when γ_F is bounded: the sphere-area test [5, Thm. 7.5].

(e) (Thm. 6.4.4) If a set Ω of directions of positive surface measure has $\sup_{\varphi\in\Omega} \int_1^\infty \mathrm{d}r/(r^{m-1} f^*(r\varphi)) < \infty$, then δ_F is inadmissible.

(f) (Thm. 6.4.2) Under $|\gamma_F(x)| \leq k/|x|$ the test in (d) is an equivalence.

Brown's proofs are those of the geometric statements: (a) is his energy integral and variational problem, (d) is the radial test function built from $1/(r^{m-1}\bar{f})$, and (e) integrates the one-dimensional capacity bound along rays. Table 1 collects the dictionary.

Corollary 6 ([IDENTITY], check C24)**.** *The ball-volume test* $\int^\infty \rho\,\mathrm{d}\rho/\mu(B_\rho) = \infty$ *[5, Thm. 7.3] follows from Theorem 5(d) by the Cauchy–Schwarz inequality on dyadic shells, and is strictly weaker.*

Remark 7 ([OBSERVATION]: priority). Grigor'yan [5, p. 178] attributes the sphere-area test for Riemannian manifolds to Lyons–Sullivan [15] and Grigor'yan [16]. Brown's Theorem 6.4.3 (1971) is the same test on W_F, and his Theorem 4.3.1 is the capacity characterization of parabolicity. In the sources we read neither literature cites the other. The analytic content of Theorem 5(a)–(b), admissibility as recurrence of the Dirichlet

Table 1: Dictionary, with status. Theorem numbers refer to Brown [1], Komaki [2] and Grigor'yan [5].

statistics	weighted Riemannian geometry	status
model manifold with Fisher metric (M, g)	Riemannian manifold	definition [10]
prior $f = e^{-\Phi}\pi_J$	smooth metric measure space $(M, g, e^{-\Phi}\,\mathrm{d}V)$	definition
$O(N^{-2})$ risk reduction over Jeffreys	$(R^{\mathrm{m}}_{\Phi} - R)/(2N^2) = -2\Delta u/(N^2 u)$	identity (Prop. 1)
asymptotic domination over Jeffreys (Komaki Thm. 2)	u nonconstant positive superharmonic	theorem
a shrinkage prior exists (Komaki Thm. 1)	(M, g) non-parabolic	theorem
prior-independent $O(N^{-2})$ risk	embedding curvatures, skewness, fourth moment; **not** R	negative (Prop. 2)
change of vague prior h (Komaki §3)	conformal change $\tilde g = (h/\pi_J)^{2/(d-2)} g$	theorem (Prop. 4)
Stein prior in Komaki's frame	flat cylinder $S^{d-1} \times \mathbb{R}$, origin at infinity	theorem (Komaki Ex. 1)
δ_F admissible, bounded risk (Brown Thm. 5.1.1)	$W_F = (\mathbb{R}^m, \delta, f^*\,\mathrm{d}x)$ parabolic	theorem
Brown's energy infimum (Thm. 4.3.1)	capacity of the unit ball in W_F (Grig. Thm. 5.1)	theorem
Brown Thm. 6.3.1 (spherical F)	model-manifold test $\int \mathrm{d}r/S(r)$ (Grig. Prop. 3.1)	theorem (C23)
Brown Thm. 6.4.3 (any F)	sphere-area parabolicity test (Grig. Thm. 7.5)	theorem; Remark 7
—	ball-volume test $\int \rho\,\mathrm{d}\rho/V(\rho) = \infty$ (Grig. Thm. 7.3)	corollary of 6.4.3, weaker (C24)
MLE admissible iff $m \le 2$ (Stein [14])	Brownian motion recurrent iff $m \le 2$	theorem ($f^* \equiv 1$)
Bakry–Émery flow $\partial_t g = -2(\mathrm{Ric} + \nabla\nabla\Phi)$	gradient flow of $\mathcal{F}$ at fixed measure	theorem (Perelman), no statistical process (§6)

form $\int |\nabla j|^2 f^*\,\mathrm{d}x$, is well established in statistics [1, 17, 18, 9]; its geometric reading was not found in the literature we searched.

5 The reach of Brown's radial tests

Theorem 5(d) is a recurrence test and (e) a transience test; Brown notes [1, p. 900] that weights exist on which both are silent. The next two lemmas say where.

Lemma 8 (spherical A_2 makes the radial tests exhaustive)**.** *Suppose* $\sup_{r\ge 1} Q(r) = C < \infty$. *If* $\int_1^\infty \mathrm{d}r/(r^{m-1}\bar f(r)) = \infty$, *then* W_F *is parabolic and* δ_F *is admissible whenever* γ_F *is bounded. If the integral is finite,* δ_F *is inadmissible.*

Proof. The first case is Theorem 5(d). In the second, let $R(\varphi) = \int_1^\infty \mathrm{d}r/(r^{m-1} f^*(r\varphi))$. By Fubini its mean over the unit sphere is $\int_1^\infty A(r)\,\mathrm{d}r/r^{m-1} \le C\int_1^\infty \mathrm{d}r/(r^{m-1}\bar f(r)) < \infty$, so by Chebyshev the set $\Omega = \{\varphi : R(\varphi) \le 2\,\mathrm{mean}\,R\}$ has at least half the surface measure and R is bounded on it. This is the hypothesis of Theorem 5(e). □

Brown's own remark after his Theorem 6.4.4 already contains the Fubini form of (e) and the Jensen inequality $A(r) \ge 1/\bar f(r)$; the lemma only observes that a bounded ratio closes the gap. Consequently every weight on which both tests are silent has $Q(r) \to \infty$ along some sequence: unbounded contrast between the arithmetic and harmonic sphere means of f^*.

Lemma 9 (doubling and Poincaré force A_2 on centred intervals)**.** *Let* $w > 0$ *be continuous on* $\mathbb{R}$, $\mu = w\,\mathrm{d}t$, $\nu = w^{-1}\,\mathrm{d}t$. *Suppose* μ *is doubling,* $\mu(2I) \le D\,\mu(I)$ *for every interval* I, *and satisfies the scale-invariant Poincaré inequality* $\int_I |u - u_I|^2\,\mathrm{d}\mu \le C|I|^2 \int_I |u'|^2\,\mathrm{d}\mu$ *for all* I *and smooth* u. *Then* $\mu([-r,r])\,\nu([-r,r]) \le 32\,C D^3\,r^2$ *for every* $r > 0$.

Proof. Take $I = [-2r, 2r]$ and the test function $u = 0$ on $[-2r, -r]$, $u = 1$ on $[r, 2r]$, and $u(t) = \nu([-r,t])/\nu([-r,r])$ in between, which

Table 2: Five one-dimensional weights $w(x_1)$ on $\mathbb{R}^3$, from the check scripts (values at $r = 10^8$; "$\propto$" marks growth with r). "Poincaré" is the Muckenhoupt–Hardy bound $4 \min_\xi \max(B_r, B_l)/|I|^2$ on the scale-invariant Poincaré constant [19]; "fails" is certified by the test function of Lemma 9. Tests (d) and (e) are those of Theorem 5.

$w(t)$	doubling	Poincaré	$Q(10^8)$	test (d)	test (e)	verdict
$(1+\|t\|)^{-1/2}$	2.7, bounded	0.47	1.33	converges	bounded	inadmissible by (e)
$(1+\|t\|)^{-1}$	17, $\propto \log r$	0.48	9.2, $\propto \log r$	diverges	infinite	admissible by (d)
$(1+\|t\|)^{-1}\log(e+\|t\|)$	9.2, $\propto \log r$	0.49	4.8, $\propto \log r$	converges	infinite	undecided; inadmissible (Ex. 12)
$(1+\|t\|)^{2}$	≤ 8, bounded	fails, $\propto r$	3×10^7, $\propto r$	converges	bounded	inadmissible by (e)
$(1+\|t\|)^{-2}$	$\propto r$	0.44	3×10^7, $\propto r$	diverges	infinite	admissible by (d)

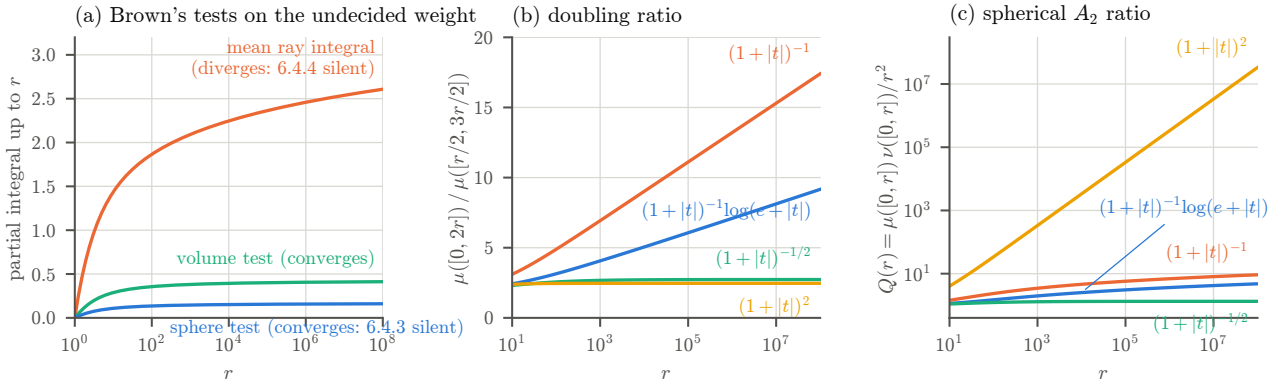


Figure 2: (a) Brown's radial tests on the weight of Example 12: the sphere-average integral converges (Theorem 6.4.3 silent), the direction-averaged ray integral diverges like $\log\log r$ (Theorem 6.4.4 silent), and the volume integral converges. (b) The doubling ratio $\mu([0,2r])/\mu([r/2,3r/2])$ of four weights: bounded for $(1+|t|)^{-1/2}$ and $(1+|t|)^2$, growing like $\log r$ for the two borderline weights, which are therefore not doubling. (c) The spherical A_2 ratio $Q(r)$: bounded only for $(1+|t|)^{-1/2}$, the one weight that is both doubling and Poincaré (Lemma 9); $(1+|t|)^2$ is doubling but fails the Poincaré inequality. All curves are recomputed from the scripts of record.

minimizes the energy across the middle segment: $\int_I |u'|^2 \,\mathrm{d}\mu = 1/\nu([-r,r])$. A function equal to 0 on a set of measure a and to 1 on a set of measure b has weighted variance at least $ab/(a+b) \geq \frac{1}{2}\min(a,b)$. Since $[-r,r]$ is contained in the eightfold dilates of $[r,2r]$ and of $[-2r,-r]$, doubling gives $\mu([r,2r]) \geq \mu([-r,r])/D^3$, and likewise for $\mu([-2r,-r])$. The Poincaré inequality now reads $\mu([-r,r])/(2D^3) \leq C(4r)^2/\nu([-r,r])$. □

Both hypotheses are needed (Table 2, Fig. 2b,c): $(1+|t|)^2$ is doubling but fails the Poincaré inequality, because the valley at 0 is a bottleneck between two heavy sides, and its ratio grows like $r/3$; $(1+|t|)^{-2}$ satisfies the Poincaré inequality but is not doubling, and its ratio also grows like $r/3$.

Corollary 10 (product weights)**.** *Let $m \geq 3$ and let $f^*(x) = w(x_1)$, or its Gaussian smoothing, with w as in Lemma 9. Then $Q(r)$ is bounded, so Lemma 8 applies; in fact δ_F is inadmissible by Theorem 5(e), and the volume-growth criterion of Remark 11 decides nothing that Brown's two tests do not.*

Proof. The sphere means of $w(x_1)$ are weighted means of w over $[-r,r]$ with a bounded kernel that is bounded below on $[-r/\sqrt{2}, r/\sqrt{2}]$, so by doubling $\bar{f}(r) \asymp \mu([-r,r])/r$ and $A(r) \lesssim \nu([-r,r])/r$; Lemma 9 bounds Q. Doubling also gives $\mu([-r,r]) \geq c\, r^{\varepsilon}$ for some $\varepsilon > 0$ (reverse doubling), so the sphere integral $\int^\infty \mathrm{d}r/(r^{m-1}\bar{f})$ is dominated by $\int^\infty r^{2-m-\varepsilon}\,\mathrm{d}r < \infty$ for $m \geq 3$, and the second case of Lemma 8 applies. Gaussian smoothing changes w by a bounded factor when $\log w$ has a bounded derivative, and changes nothing above. □

Remark 11 (the volume criterion, conditional). Under doubling and a scale-invariant Poincaré inequality on balls, the Dirichlet form of W_F satisfies the parabolic Harnack inequality and two-sided Gaussian heat-kernel bounds [20, 21, 22] (Muckenhoupt A_2 weights are an example [23]), and then transience is equivalent to

$\int^\infty r\,\mathrm{d}r/\mu(B(0,r)) < \infty$ [5, Thm. 11.1 and its proof]. Combined with Theorem 5(b) this would be an admissibility criterion for W_F. Corollary 10 shows it is empty of new content for product weights. Whether a non-product doubling–Poincaré weight exists on which both of Brown's tests are silent is open: the one-dimensional mechanism is excluded by Lemma 9, but in $\mathbb{R}^m$ an escape route that is cheap without being radial is not excluded by our argument, and we found no example. An earlier internal record of this gate asserted the bound of Lemma 9 for the whole class without proof; that step is withdrawn and replaced by the lemma.

Example 12 (an undecided weight, decided)**.** Let $F(\,\mathrm{d}\theta) = (1+|\theta_1|)^{-1}\log(e+|\theta_1|)\,\mathrm{d}\theta_1\,\mathrm{d}\theta_2\,\mathrm{d}\theta_3$ on $\mathbb{R}^3$, so that f^* depends on x_1 only, γ_F is bounded, and $f^* \asymp (1+|x_1|)^{-1}\log(e+|x_1|)$. Theorem 5(d) is silent: $\bar{f}(r) \asymp \log^2 r/r$ and the sphere integral converges. Theorem 5(e) is silent: along a direction with $\cos\varphi \neq 0$ the ray integral is $|\cos\varphi|\int^\infty \mathrm{d}r/(r\log r) = \infty$, and the equatorial directions have measure zero; (f) does not apply since $\gamma_F \asymp 1/|x_1|$ is not $O(1/|x|)$. The weight is not doubling either, its doubling ratio grows like $\log r$ (Fig. 2b), so Remark 11 does not apply, although its integral happens to converge. The product structure decides: $\{Z_t\}$ is the product of the one-dimensional diffusion W with generator $w^{-1}(wu')'$ and a Brownian motion on $\mathbb{R}^2$, and $\{Z_t\}$ is transient iff $\int^\infty p_t^W(0,0)\,(4\pi t)^{-1}\,\mathrm{d}t < \infty$. A Crank–Nicolson computation of $p_t^W(0,0)$ (density with respect to $w\,\mathrm{d}t$; the control $w \equiv 1$ reproduces $(4\pi t)^{-1/2}$ to five digits) gives $\log^2 t \cdot p_t^W(0,0) = 2.58, 2.95, 3.18, 3.33, 3.44$ at $t = 10^3, \ldots, 10^7$, i.e. $p_t^W(0,0) \asymp 1/\log^2 t$, as the null-recurrent heuristic $p_t^W(0,0) \approx 1/\mu([-\sqrt{t},\sqrt{t}])$ predicts. Hence the integral converges, $\{Z_t\}$ is transient and δ_F is inadmissible by Theorem 5(b). A rigorous citation for the one-dimensional asymptotic is owed before the example is used elsewhere. The method is the one with which Johnstone and Lalley [24] answered Brown's question (his §6.5) whether any non-proper-Bayes estimator is immune to the Stein effect under products; the example is an instance of their phenomenon: a recurrent one-dimensional generalized Bayes piece whose product with the admissible two-dimensional MLE is inadmissible.

6 Four geometries, no horizon, no dynamics

The phrase "interior geometry" in the blueprint denoted four different objects, and every one of its physical claims came from conflating them. (1) The Fisher–Rao metric of the model: flat for the normal location family, hyperbolic for a location–scale family (Komaki's Example 2); for $N(\mu,\sigma^2)$, $g = \mathrm{diag}(1/\sigma^2, 2/\sigma^2)$ with curvature $-1/2$ (check C22). This is the horn of Fig. 1a: the throat is $\sigma \to \infty$, the flare $\sigma \to 0$, and the axial coordinate $-\ln\sigma \approx \frac{1}{2}\ln N$ is an observation count, an *external* clock. Its curvature is the model's; the curvature in §§3–4 is the prior's. (2) The weighted manifold $(M, g, e^{-\Phi}\,\mathrm{d}V)$, where Propositions 1–2 and Theorem 5 live. (3) Komaki's conformal frame, Fig. 1b, in which the Stein prior becomes a flat cylinder with the origin at infinite distance. (4) The pullback $J^\top\Sigma^{-1}J$ of the estimator map (check C11), a legitimate Riemannian object on the data space whose angular part degenerates at $\|x\| = \sigma\sqrt{d-2}$ for the James–Stein map, where its tangential Jacobian eigenvalue vanishes and the radial one equals 2 (check C12; Fig. 1c). This is the only "horizon" in sight, and it is a statement about an estimator, not about causal structure; no Schwarzschild form arises (check C3).

There is no dynamics. In Perelman's $\mathcal{F}$ the measure is a gauge and the metric flows; in statistics the Fisher metric is fixed by the model and all content sits in the measure, which Bayesian updating moves at fixed g through $f_N \propto f\,L_N$, a log-likelihood increment and not a Ricci-type flow. The Bakry–Émery flow of Table 1 therefore has no statistical process generating it as the framework stands, and any future dynamical claim must supply a mechanism that changes g and must declare its clock.

7 Assessment and record

What this note can claim: the dictionary of Table 1; Proposition 1 with its sign; Proposition 2 as the reason the dictionary stops at the gradient term; Theorem 5 as the geometric reading of a formulation that statistics already owns in Dirichlet-form language; Remark 7 as a historical observation; Lemmas 8–9 as corollaries of Brown's theorems. What it cannot claim: any gravitational or

dynamical statement; the words "horizon", "Einstein frame" or "spacetime" except as names of functional forms; any novelty wording before a citation-index sweep. The owner's assessment of record is that nothing here rises to a contribution; the research question that the dictionary left behind, a new admissibility criterion from weighted potential theory, closed on its first serious test (§5) with one small open residue (Remark 11). The stream is closed and this note is its record.

Verification record. Equation lint: 22 symbolic and Monte Carlo checks (`checks/lint_admissible_spacetime.py`; 6 FAIL on the blueprint's claims, 12 PASS, 4 INFO). Bibliography: 24 references of the blueprint checked field by field against publisher, Crossref or arXiv records (22 verified, 1 with corrections, 1 unverified). Numerics: `checks/g0_feasibility.py` (sphere, ray and volume integrals; Crank–Nicolson heat kernel) and `checks/g0_doubling_poincare.py` (doubling ratios, Muckenhoupt–Hardy bounds, spherical ratios); every number in Table 2 and Fig. 2 is produced by these scripts. Brown [1] was read from a scan of the journal pages; its theorems are cited by their numbers there.

References

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